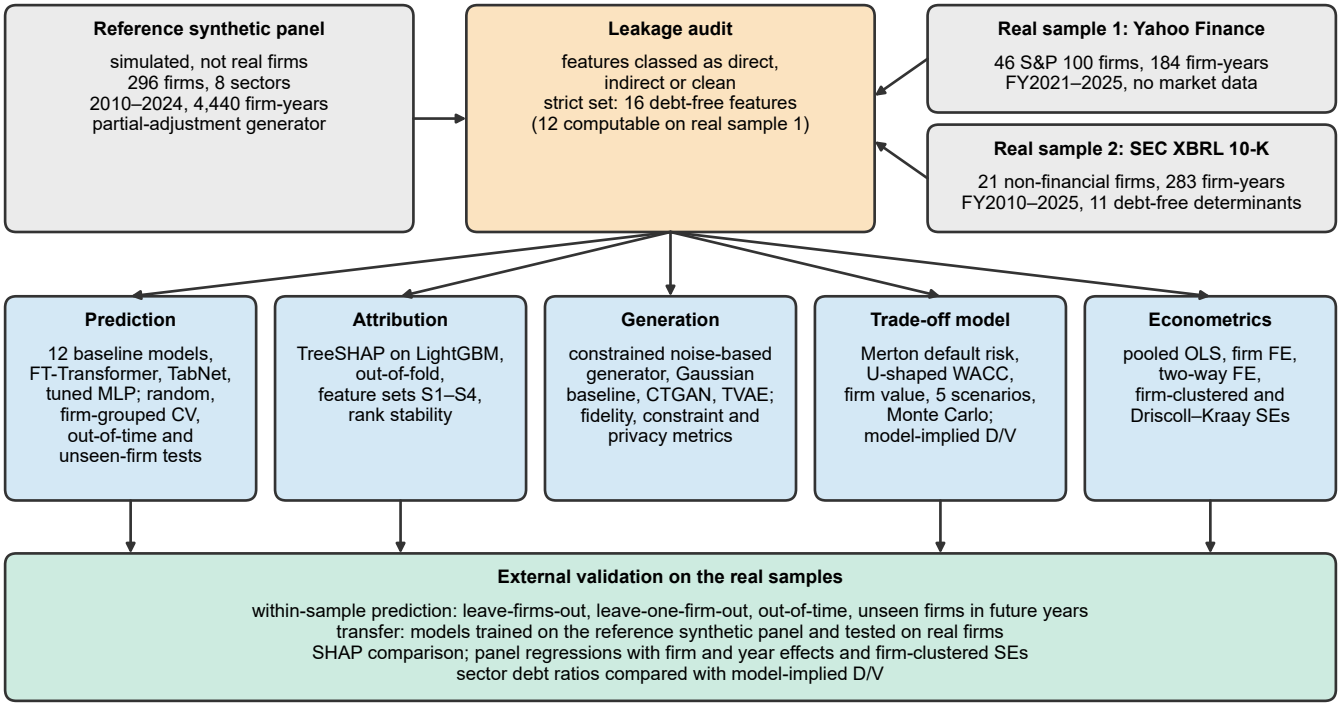

Reference synthetic panel
simulated, not real firms
296 firms, 8 sectors
2010–2024, 4,440 firm-years
partial-adjustment generator
Leakage audit
features classed as direct,
indirect or clean
strict set: 16 debt-free features
(12 computable on real sample 1)
Real sample 1: Yahoo Finance
46 S&P 100 firms, 184 firm-years
FY2021–2025, no market data
Real sample 2: SEC XBRL 10-K
21 non-financial firms, 283 firm-years
FY2010–2025, 11 debt-free determinants
Prediction
12 baseline models,
FT-Transformer, TabNet,
tuned MLP; random,
firm-grouped CV,
out-of-time and
unseen-firm tests
Attribution
TreeSHAP on LightGBM,
out-of-fold,
feature sets S1–S4,
rank stability
Generation
constrained noise-based
generator, Gaussian
baseline, CTGAN, TVAE;
fidelity, constraint and
privacy metrics
Trade-off model
Merton default risk,
U-shaped WACC,
firm value, 5 scenarios,
Monte Carlo;
model-implied D/V
Econometrics
pooled OLS, firm FE,
two-way FE,
firm-clustered and
Driscoll–Kraay SEs
External validation on the real samples
within-sample prediction: leave-firms-out, leave-one-firm-out, out-of-time, unseen firms in future years
transfer: models trained on the reference synthetic panel and tested on real firms
SHAP comparison; panel regressions with firm and year effects and firm-clustered SEs
sector debt ratios compared with model-implied D/V